

1	2	3	4	5	6	Sum
/20	/20	/20	/20	/20	/20	/100

Name _____

Jan 14, 2018 Sunday

SID _____

School/Department _____

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu) = \int_0^{+\infty} e^{-t} t^{\nu-1} dt, \quad \Gamma(\nu+1) = \nu\Gamma(\nu), \quad \Gamma(1) = 1, \quad \Gamma(1/2) = \sqrt{\pi}; \quad \int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}.$$

c	h	$\hbar$	e	m_e	m_n	u	N_A	k_B
3.0×10^8 m/s	6.63×10^{-34} J·s 1.24×10^3 eV·nm/c	1.05×10^{-34} J·s	1.6×10^{-19} C	9.11×10^{-31} kg $0.511 \text{ MeV}/c^2$	939.565 MeV/c^2	931.494 MeV/c^2	6.02×10^{23} mol^{-1}	8.617×10^{-5} eV/K

Quantum numbers of three quarks:

Quark symbol	Charge number	Spin	Baryon number	Strangeness
u	2/3	1/2	1/3	0
d	-1/3	1/2	1/3	0
s	-1/3	1/2	1/3	-1

Select five out of the following six problems.

1.(20 pts.) An X-ray photon of wavelength $\lambda_0 = 0.0300$ nm makes a collision with an electron that is initially at rest and undergoes a 90° Compton scattering.

- Find the wavelength of scattered photon.
- Find the kinetic energy of the electron after the scattering.

2. (20 pts.) The half-life of neutron is $t_{1/2} = 10.2$ min. A beam of neutrons of kinetic energy 5 eV is emitted in a direction.

- What is the de Broglie wavelength of the neutrons?
- Over what distance will the intensity of the neutron beam be reduced to the half of its original value?
- What is the average distance a neutron in the beam travels before it decays?

3. (20 pts.) A particle of mass m is moving in a one-dimensional simple harmonic potential and its natural circular frequency is ω . The particle is in a stationary quantum state of the normalized wave function

$$\psi(x) = C e^{-\frac{1}{2}\alpha^2 x^2}$$

where $\alpha = \sqrt{m\omega/\hbar}$ and C is a positive constant.

(a) Write the Hamiltonian operator of the particle. Determine the constant C .

(b) Is the wave function an eigenfunction of the Hamiltonian?

(c) Find the averages $\langle xp \rangle$ and $\langle px \rangle$ in the quantum state, where x and p are position and momentum of the particle.

4. (20 pts.) A neutron star that is composed almost entirely of N neutrons can be approximately treated as ideal Fermi gas of uniform density in a sphere of radius R at temperature $T = 0$.

- (a) Find the Fermi energy of the neutron gas in terms of neutron number N , radius of the star R and the mass of neutron m_n .
- (b) Find the total kinetic energy of the neutron gas E_K .
- (c) The gravitational energy of the neutron star can be written as

$$E_G = -\frac{3}{5} \frac{GM^2}{R} = -\frac{3}{5} \frac{GN^2 m_n^2}{R}.$$

The star is in equilibrium when the inward gravitational force is just opposed by the repulsion due to the Pauli principle, that is, when the total energy $E = E_K + E_G$ reaches a minimum value. Show that the radius of the neutron star is given by

$$R = \frac{\hbar^2}{GN^{1/3} m_n^3} \left(\frac{9\pi}{4} \right)^{\frac{2}{3}} = \frac{\hbar^2}{GN^{1/3} m_n^3} \left(\frac{9}{32\pi^2} \right)^{\frac{2}{3}}.$$

5. (20 pts.) Consider a one-dimensional potential step

$$U(x) = U_0 \theta(x) = \begin{cases} 0, & x < 0, \\ U_0, & x \geq 0, \end{cases} \quad (U_0 > 0)$$

and incident particles of mass m and energy $E < U_0$ coming from left ($x < 0$) to collide with the step. The solutions of the wave equations can be written as

$$\begin{aligned} \psi(x) &= Ae^{ikx} + B^{-ikx} \quad \text{for } x < 0; \\ \psi(x) &= Ce^{-\kappa x} \quad \text{for } x \geq 0 \end{aligned}$$

where $k = \sqrt{2mE}/\hbar$ and $\kappa = \sqrt{2m(U_0 - E)}/\hbar$.

- (a) What is the meaning of the each term of the solutions above?
- (b) By applying the boundary conditions, determine the ratios B/A and C/A .
- (c) Find the reflection coefficient R and the transmission coefficient T of the particle.

6.(20 pts.) (a) For the hadrons and their quark combinations listed in the first column of the following table, classify them as baryon and meson, find the charge number, baryon number and strangeness of them and write the results into the table.

Particle and quark combination	Classification	Baryon number	Charge number	Strangeness
$p - uud$				
$\Lambda^0 - uds$				
$K^- - s\bar{u}$				
$\pi^+ - u\bar{d}$				
$\Omega^- - sss$				
$K^0 - d\bar{s}$				
$n - udd$				
$K^+ - u\bar{s}$				

(b) Analyze following reactions. Identify them as “allowed” or “forbidden”. If the reaction is allowed, indicate what interaction it is mainly mediated. If the reaction is forbidden, name the conservation law(s) violated.

Reactions or decays	Allowed/Forbidden	Main interaction/violated conservation law(s)
$p + p \rightarrow p + \pi^+ + \Lambda^0 + K^0$		
$K^- + p \rightarrow n + \Lambda^0$		
$\nu_e + p \rightarrow n + e^+$		
$\Omega^- \rightarrow \Lambda^0 + K^-$		
$p + p \rightarrow p + n + K^+$		

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Jan 11, 2019

Friday

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University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

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c	h	$\hbar$	e	m_e	m_n	u	N_A	k_B
3.0×10^8 m/s	6.63×10^{-34} J·s 1.24×10^3 eV·nm/c	1.05×10^{-34} J·s	1.6×10^{-19} C	9.11×10^{-31} kg $0.511 \text{ MeV}/c^2$	939.565 MeV/c^2	931.494 MeV/c^2	6.02×10^{23} mol^{-1}	8.617×10^{-5} eV/K

Quantum numbers of three quarks:

Quark symbol	Charge number	Spin	Baryon number	Strangeness
u	2/3	1/2	1/3	0
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Select five out of the following six problems.

1.(20 pts.) In the experiment of photoelectric effect using iron as emitter, it is found that the threshold frequency of light is $\nu_C = 1.10 \times 10^{15}$ Hz.

(a) Find the work function ϕ of iron in the unit of eV.

(b) Find the stopping voltage V_0 for the photoelectrons emitted when light of wave length $\lambda = 250 \text{ nm}$ falls on the surface of iron.

2. (20 pts.) In the beta decay of ${}^{24}_{11}\text{Na}$, the daughter nucleus is ${}^{24}_{12}\text{Mg}$ and an electron with kinetic energy of 2.15 MeV is observed. The atomic mass of the ${}^{24}_{11}\text{Na}$ is $m({}^{24}_{11}\text{Na}) = 23.990963 \text{ u}$ and the atomic mass of ${}^{24}_{12}\text{Mg}$ is $m({}^{24}_{12}\text{Mg}) = 23.985042 \text{ u}$.

(a) Write down the nuclear equation of the decay.

(b) Find the Q-value of the decay.

(c) What is the energy of the accompanying neutrino/antineutrino?

3. (20 pts.) For each of the following reactions, identify it is allowed or forbidden. If it is forbidden, state a conservation law that is violated.

(a) $p + p \rightarrow p + \Lambda^0 + K^+$

(b) $K^0 \rightarrow \pi^+ + e^-$

(c) $p + n \rightarrow p + \Lambda^0$

(d) $\Omega^- \rightarrow \Lambda^0 + K^-$

(e) $n + \nu_e \rightarrow p + e^-$

(f) $\Lambda^0 \rightarrow p + \pi^-$

(g) $p + \bar{p} \rightarrow \Lambda^0 + \Lambda^0$

The quark combinations of relevant hadrons:

$$p - uud \quad \Lambda^0 - uds \quad \Omega^- - sss \quad \bar{p} - \bar{u}\bar{u}\bar{d} \quad K^+ - u\bar{s} \quad K^0 - d\bar{s} \quad K^- - s\bar{u} \quad \pi^+ - u\bar{d} \quad \pi^- - d\bar{u}$$

4. (20 pts.) An ideal Bose or Fermi gas is composed of particles whose energy-momentum relation is $\varepsilon = p^2/(2m)$. The distribution function, i.e., the average number of particle in a quantum state with energy ε , can be written as $f(\varepsilon) = 1/(e^{\beta(\varepsilon-\mu)} \pm 1)$ where $\beta = 1/k_B T$, μ is the chemical potential of the gas, and the positive and negative signs are for Fermi and Bose gas respectively. However, if $e^{-\beta\mu} \gg 1$, the ± 1 in the denominator is negligible and the quantum distributions both reduce to the classical limit of Boltzmann distribution $f_C(\varepsilon) = e^{-\beta(\varepsilon-\mu)}$.

(1) Show that the density of states (DOS) of the gas can be expressed as

$$\rho(\varepsilon) = \frac{g_s V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \varepsilon^{1/2}$$

where g_s is the degree of degeneracy due to the spin of the particle and V is the volume of the gas.

(2) Show that total number of particles N in the classical Boltzmann limit can be expressed as

$$N = g_s V e^{\beta\mu} \left(\frac{2\pi m}{\beta h^2} \right)^{3/2}.$$

(3) Furthermore, show that the condition for the classical limit (or the non-degenerate limit) $e^{-\beta\mu} \gg 1$ is equivalent to the condition $n\lambda^3 \ll g_s$ where $n = N/V$ is the number density of particles and $\lambda = h/\sqrt{2\pi m k_B T}$ is the thermal de Broglie wavelength.

5. (20 pts.) An electron of mass m moves in a one-dimensional symmetric infinite potential well

$$U(x) = \begin{cases} 0, & |x| \leq a/2, \\ +\infty, & |x| > a/2, \end{cases}$$

where $a > 0$ is the width of the well.

(a) Write down the stationary Schrödinger equation of the electron.

(b) Assuming that the solutions of the stationary Schrödinger equation take the form

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ika}, & |x| \leq a/2, \\ 0, & \text{elsewhere} \end{cases}, \quad k > 0, \quad A > 0,$$

determine the possible values of wave vector k by applying the proper boundary conditions.

(c) Using the results of (b) to find the energy levels of the electron in the potential well.

(d) Determine the coefficients A and B in (b) by the normalization of the wave functions.

6.(20 pts.) The Hamiltonian operator of a one-dimensional harmonic oscillator is $\hat{H} = \hat{p}^2 / (2m) + (1/2)m\omega^2 x^2$ where m is the mass of the particle, $\hat{p} = -i\hbar(d/dx)$ is the momentum operator and ω is the natural circular frequency of the oscillator. The normalized ground state wave function of the oscillator is found to be

$$\psi_0(x) = \left(\frac{\alpha}{\sqrt{\pi}} \right)^{1/2} e^{-\frac{\alpha^2 x^2}{2}}, \quad \alpha = \left(\frac{m\omega}{\hbar} \right)^{1/2}.$$

- (a) Find the expectation values of position coordinator $\langle x \rangle$ and momentum $\langle p \rangle$ of the particle in the ground state.
- (b) Find the uncertainties of position $\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$ and momentum $\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2}$, and show that the Heisenberg's uncertainty principle is satisfied.
- (c) Find the expectation values of kinetic energy and potential energy of the oscillator in its ground state. Comment on their non-zero values and relative magnitude of two quantities.

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Name _____

January 10, 2020

Friday

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University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

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Quantum numbers of three quarks:

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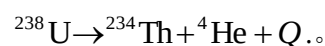
Select five out of the following six problems.

1.(20 pts.) If the electron of a hydrogen atom is replaced by a tau (τ) lepton, a hydrogen tau atom is formed. The rest mass of tau lepton is $m_\tau = 1777.05 \text{ MeV}/c^2$.

(a) Find the reduced mass for the relative motion of the tau (τ) lepton and the proton in the hydrogen tau atom.

(b) By applying the Bohr's quantization rule, find the first Bohr's radius a_0 and the energy levels E_n of the hydrogen tau atom.

2. (20 pts.) The α decay process of a ^{238}U (Uranium-238) nucleus is



It is known that the decay energy (Q – value) of the process is 4.27 MeV . Here we assume that mother nucleus ^{238}U is initially at rest.

(a) Find the kinetic energy of the emitting α particle (^4He (Helium-4) nucleus).

(b) Find the kinetic energy of the recoiling daughter ^{234}Th (Thorium-234) nucleus.

3. (20 pts.) For each of the following reactions, identify it as allowed or forbidden. If it is forbidden, state at least one conservation law that is violated. If it is allowed, state the main interaction the reaction is mediated.

	allowed/forbidden	violated conservation law/main interaction
(a) $p + p \rightarrow p + p + K^+ + K^-$	(a)	
(b) $p + K^- \rightarrow \Sigma^+ + n + \pi^+ + \pi^-$	(b)	
(c) $\pi^+ + p \rightarrow K^+ + \Sigma^+$	(c)	
(d) $\bar{\nu}_\mu + p \rightarrow \mu^+ + n$	(d)	
(e) $p + p \rightarrow p + n + K^+$	(e)	
(f) $K^+ \rightarrow \pi^+ + \pi^0$	(f)	
(g) $K^+ \rightarrow \mu^+ + \nu_\mu$	(g)	

The quark combinations of relevant hadrons:

$p - uud$ $n - udd$ $\Sigma^+ - uus$ $K^+ - u\bar{s}$ $K^- - s\bar{u}$ $\pi^+ - u\bar{d}$ $\pi^- - d\bar{u}$

4. (20 pts.) The electronic stationary wave function of a hydrogen atom at its ground state can be written in the form

$$\psi(r) = Ce^{-r/a_0}$$

where r is the radial distance in the spherical coordinates, a_0 the Bohr radius and C the normalization coefficient to be determined.

- Determine the normalization coefficient C in terms of a_0 .
- Find the corresponding radial probability density $w(r)$.
- Calculate the most probable radius $r_{\max}$ and the average radius $\langle r \rangle$ of the electron.

5. (20 pts.) A particle of mass m moves in a one-dimensional infinite potential well

$$U(x) = \begin{cases} 0, & 0 < x < a, \\ +\infty, & x \leq 0 \text{ or } x \geq a, \end{cases}$$

where $a > 0$ is the width of the well. There is a normalized wave function

$$\psi(x, t) = \begin{cases} \frac{3}{5} \sqrt{\frac{2}{a}} \sin(k_1 x) e^{-i\omega_1 t} + A \sqrt{\frac{2}{a}} \sin(k_2 x) e^{-i\omega_2 t}, & 0 < x < a, \\ 0, & x \leq 0 \text{ or } x \geq a, \end{cases}$$

where $k_1 = \pi/a$, $k_2 = 2\pi/a$, $\omega_1 = \hbar k_1^2 / (2m)$, $\omega_2 = \hbar k_2^2 / (2m)$ and A is a constant to be determined.

- Verify that the wave function satisfies the Schrodinger equation and the boundary conditions for the potential well.
- Determine the constant A .
- Find the corresponding probability density $\rho(x, t)$ of the particle.
- Find the average position $\langle x \rangle$ of the particle as the function of time t .

- 6.(20 pts.) Consider a gas of N extremely relativistic spin-1/2 fermions that are confined in a volume V . The energy-momentum relation of the fermions is $\varepsilon = pc$.
- (a) Find the density of states (i.e., the number of quantum states per unit energy interval) of the system.
 - (b) Find the Fermi energy E_F of the system.
 - (c) Show that the total energy of the system at $T = 0$ K is $(3/4)NE_F$.